Literature Review & Novelty Analysis

Cloud-Native Hybrid Threat Detection System with Agentic AI

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1. Introduction

This document presents a structured literature review of the most recent research papers (2025–2026) that are directly relevant to the Cloud-Native Hybrid Threat Detection System. It compares the approaches taken in current academic literature with the techniques implemented in this project, and clearly articulates the **novelty** of this work.

The project integrates four key research domains:

1. Anomaly Detection using Isolation Forest
2. Hybrid IDS/UEBA Fusion Architectures
3. Agentic AI with ReAct Reasoning for Cybersecurity
4. Explainable AI (XAI) and Alert Fatigue Reduction

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2. Literature Review — Paper-by-Paper Analysis

2.1 Agentic AI and ReAct Reasoning for Cybersecurity

Paper 1: "Autonomous Cyber Incident Response Using Reasoning and Action"

- **Authors:** Sudipto Baral, Sajal Saha, Anwar Haque
- **Published:** 2025, IEEE International Wireless Communications and Mobile Computing Conference (IWCMC 2025)

Summary: This paper presents a framework using LangGraph and ReAct agents to automate response to cybersecurity threats like SSH brute-force and botnet intrusions. The system uses a LangGraph-based stateful loop where the LLM reasons about the threat, calls tools (firewall rules, endpoint isolation), observes the result, and iterates until the incident is resolved.

Comparison with Our Project: Both our system and Baral et al.'s work implement the ReAct paradigm for autonomous cyber response. However, there are critical differences:

- **Their approach** relies on LangGraph (a cloud-dependent framework) and cloud-hosted LLMs, incurring API costs and privacy risks.
- **Our approach** runs entirely locally using Ollama (Qwen 2.5 / Llama 3), achieving zero cost and complete data privacy.
- **Their system** focuses on post-incident response (SSH brute-force, botnets already detected).
- **Our system** covers the full pipeline: detection (IDS + UEBA) → fusion (60/40 algorithm) → AI reasoning → autonomous response, making it an end-to-end solution.

Paper 2: "Breaking ReAct Agents: Foot-in-the-Door Attack Will Get You In"

- **Authors:** Itay Nakash, George Kour, Guy Uziel, Ateret Anaby-Tavor (IBM Research AI)
- **Published:** 2025, Findings of the Association for Computational Linguistics: NAACL 2025

Summary: This paper explores security vulnerabilities in ReAct agents, demonstrating how indirect prompt injection can manipulate the agent's thought process to perform malicious tool actions. The authors propose a reflection-based defense mechanism where the agent reviews its own reasoning chain before executing high-impact actions.

Comparison with Our Project: This paper highlights a risk that is relevant to our system. Our agentic threat agent uses tool-calling (check_ip_reputation, get_similar_threats, etc.), which could theoretically be vulnerable to adversarial manipulation. However, our system has built-in safeguards:

- **Tool whitelist:** Only 8 pre-defined tools are available; the AI cannot call arbitrary functions.
- **Rule-based fallback:** If the AI produces an invalid action, the system falls back to the mathematical 60/40 fusion, ensuring reliable response even if the LLM is compromised.
- **Graduated response:** The 4-tier response system (LOG/ALERT/RATE_LIMIT/BLOCK) prevents catastrophic over-blocking.

Paper 3: "LLM-Assisted Proactive Threat Intelligence for Automated Reasoning"

- **Authors:** (Multiple authors)
- **Published:** April 2025, arXiv preprint

Summary: This paper details an approach integrating LLMs with Retrieval-Augmented Generation (RAG) and continuous threat intelligence feeds to automate real-time threat detection and response. It addresses the limitations of static analysis by enabling dynamic, context-aware threat assessment using external knowledge bases.

Comparison with Our Project:

- **Their approach** uses cloud-hosted LLMs with RAG pipelines, requiring vector databases and external API calls for threat intelligence retrieval.
- **Our approach** achieves similar contextual reasoning using a simpler architecture: local Ollama inference + SQLite persistent memory + tool-calling for IP reputation checks. This trades some reasoning depth for complete privacy, zero cost, and simpler deployment.
- **Key advantage of our system:** We combine AI reasoning with a mathematical detection backbone (60/40 fusion), providing a reliable fallback that RAG-only systems lack.

Paper 4: "AI-Driven Autonomous Incident Response: Revolutionizing Cybersecurity Operations with Real-Time Threat Mitigation"

- **Authors:** Smita Verma
- **Published:** 2025, International Journal of Communication Networks and Information

Security (IJCNIS)

Summary: This paper examines AI-driven autonomous incident response for real-time threat detection and mitigation. It specifically addresses how minimizing false alarms reduces the burden on security analysts and advocates for graduated response mechanisms that scale actions proportionally to threat severity.

Comparison with Our Project: Our autonomous response agent directly implements Verma's advocated graduated response model:

- LOG (< 0.4 risk) → ALERT (0.4–0.6) → RATE_LIMIT (0.6–0.8) → BLOCK (>= 0.8)
Our contribution goes beyond Verma's theoretical framework by:
- **Actually implementing** the graduated response with real AWS Security Group modifications.
- **Adding AI explainability** so analysts understand *why* a specific response level was chosen.
- **Including a feedback loop** (outcome recording) for continuous accuracy improvement.

2.2 Isolation Forest for Anomaly Detection

Paper 5: "DeSFAM: An Adaptive eBPF and AI-Driven Framework for Securing Cloud Containers in Real Time"

- **Authors:** Sehar Zehra, Hassan Jamil Syed, Fahad Samad, Ummay Faseeha
- **Published:** 2025, IEEE Access, Vol. 13, pp. 139203–139224

Summary: DeSFAM combines eBPF-based system call monitoring with a dual detection engine: Variational Autoencoder (VAE) for deep behavioral analysis and Isolation Forest (iForest) for rapid statistical anomaly detection. The system includes contextual risk scoring mapped to the MITRE ATT&CK framework and provides real-time container security for cloud-native environments.

Comparison with Our Project: Both projects use Isolation Forest as a core detection component in cloud environments.

- **DeSFAM** focuses on container-level security (syscall monitoring via eBPF), operating at the infrastructure layer.
- **Our system** operates at the application/network layer (CloudWatch metrics + CloudTrail user activity), providing a complementary detection perspective.
- **DeSFAM** uses VAE + iForest in parallel; **our system** uses iForest × 2 (network + user) with weighted fusion.
- **Key difference:** DeSFAM lacks an AI reasoning/explainability layer. Our Agentic AI provides natural language explanations for every detection decision, which DeSFAM does not offer.

Paper 6: "siForest: Detecting Network Anomalies with Set-Structured Isolation Forest"

- **Authors:** Christie Djidjev
- **Published:** December 2025, arXiv preprint

Summary: siForest presents a variation of the Isolation Forest algorithm tailored for set-

structured data, specifically evaluated on detecting anomalies in internet scan data. Unlike standard Isolation Forest which processes individual data points, siForest treats network scans as cohesive sets, improving detection of coordinated scanning activities.

Comparison with Our Project:

- **siForest** innovates on the algorithmic level by modifying how Isolation Forest processes data structures.
- **Our project** innovates on the architectural level by using standard Isolation Forest in a novel dual-modality fusion configuration (IDS + UEBA with 60/40 weighting).
- Both approaches acknowledge that standalone Isolation Forest has limitations; siForest addresses this through algorithmic improvement, while our system addresses it through multi-source data fusion and AI augmentation.

Paper 7: "Two-Stream Hybrid Models for Network Anomaly Detection"

- **Authors:** (Multiple authors, Lviv Polytechnic National University)
- **Published:** 2025, ResearchGate / Conference Proceedings

Summary: This paper proposes a two-stream hybrid architecture where Isolation Forest serves as an initial rapid statistical filter for spatial deviations, followed by a secondary deep behavioral profiling component (Autoencoders or GANs) for analyzing complex temporal patterns. The dual-stream approach improves both detection speed and accuracy.

Comparison with Our Project: Both our system and this paper use a two-stream architecture:

- **Their streams:** Statistical filter (IF) → Deep behavioral profiler (AE/GAN) — both analyze the same data source.
- **Our streams:** Network IDS (IF on CloudWatch) → User UEBA (IF on CloudTrail) — each analyzes a fundamentally different data domain. Our approach is architecturally more novel because we fuse two *independent detection paradigms* rather than two processing stages on the same data. Additionally, we add an Agentic AI layer that their system lacks.

Paper 8: "IF-DLDD: Isolation Forest with Dynamic Concept Drift Detection"

- **Authors:** (Multiple authors)
- **Published:** 2025, IEEE Conference Proceedings

Summary: This paper presents a dynamic framework combining Isolation Forest with sliding-window concept drift detection (IF-DLDD). It uses statistical T-tests as secondary verification to validate potential drifts detected by the Isolation Forest layer, reducing false positives caused by evolving network traffic patterns without requiring constant manual retraining.

Comparison with Our Project:

- **IF-DLDD** addresses the concept drift problem (changing definition of "normal" over time) through statistical verification.
- **Our system** addresses concept drift through a different mechanism: the SQLite decision store maintains historical decisions, and the adaptive threshold system can adjust risk

boundaries based on accumulated feedback data.
- **IF-DLDD** is purely mathematical; our system adds human-in-the-loop feedback (outcome recording) and AI reasoning for more nuanced adaptation.

2.3 Hybrid IDS/UEBA and Cloud Security

Paper 9: "Hybrid AI-Based IDS with Ensemble Learning for Cloud Environments"

- **Authors:** (Multiple authors)
- **Published:** Late 2025, ResearchGate / IJIRSET

Summary: This paper proposes hybrid frameworks combining signature-based detection with ensemble learning methods (Random Forest, XGBoost) to achieve high accuracy on the CICIDS2017 benchmark dataset while maintaining low false-positive rates in cloud environments. The hybrid approach handles both known attack signatures and novel zero-day threats.

Comparison with Our Project:

- **Their approach** uses supervised ensemble learning (RF + XGBoost), requiring labeled attack datasets for training.
- **Our approach** uses unsupervised Isolation Forest, requiring no labeled data — a significant practical advantage since labeled cloud attack data is scarce and expensive to obtain.
- **Their system** performs classification (attack vs. normal); **our system** produces a continuous risk score (0–1) that maps to graduated responses, providing more nuanced threat assessment.
- **Neither their system nor any paper in this survey** combines IDS + UEBA with weighted fusion AND Agentic AI, which is our unique contribution.

Paper 10: "Adaptive IDS Combining Isolation Forest and Suricata for Zero-Day and Known Threat Detection"

- **Authors:** (Multiple authors, Universitas Jenderal Soedirman)
- **Published:** 2025, UNSOED Conference Proceedings

Summary: This paper develops an adaptive IDS that combines Isolation Forest (for zero-day/unknown threats) with Suricata (for known signature-based threats), creating a dual-layer detection system that reduces alert fatigue while maintaining comprehensive threat coverage.

Comparison with Our Project:

- **Their dual-layer:** Signature-based (Suricata) + Anomaly-based (IF) on network traffic only.
- **Our dual-layer:** Network anomaly (IF on CloudWatch) + User behavior anomaly (IF on CloudTrail) — fundamentally different data sources.
- **Key distinction:** Their fusion is at the detection method level (known vs. unknown threats); our fusion is at the data domain level (network vs. user behavior). Our approach provides contextual correlation that theirs cannot — e.g., detecting that a DDoS attack (high network risk) is external because user behavior is normal (low user risk).

2.4 Explainable AI (XAI) and Alert Fatigue Reduction

Paper 11: "SHAP-Enhanced Isolation Forest for Explainable Network Intrusion Detection"

- **Authors:** (Multiple authors)
- **Published:** 2025, Eksplorium Journal / ResearchGate

Summary: This paper integrates SHapley Additive Explanations (SHAP) with Isolation Forest models to provide feature-level explanations for network intrusion detection. Analysts receive not just anomaly scores but also insights into which specific network features (packet size, port, protocol) contributed most to the detection, increasing transparency and trust in automated security responses.

Comparison with Our Project: This paper represents the dominant XAI approach in current literature: post-hoc mathematical explanations (SHAP values).

- **Their approach:** After IF detects an anomaly, SHAP computes feature importance scores (e.g., "packet_size contributed 0.35 to the anomaly score").
- **Our approach:** The Agentic AI generates natural language reasoning in real-time (e.g., "Network risk is extremely high at 0.95 but user behavior is normal at 0.10. This pattern indicates an external volumetric attack like DDoS rather than account compromise. Recommending RATE_LIMIT.").
- **Our advantage:** Natural language explanations are immediately actionable by non-technical stakeholders and SOC analysts without requiring statistical literacy. SHAP values require interpretation expertise.

Paper 12: "Reducing False Alerts in Cybersecurity Threat Detection Using Generative AI"

- **Authors:** Xiao Lin, Glory Avina, Javier Santoyo
- **Published:** 2025, 4th Workshop on AI-Enabled Cybersecurity Analytics (affiliated with KDD)

Summary: This paper integrates Generative AI (GenAI) into threat detection to automatically suppress false alerts based on analyst feedback on historical anomalies. This creates a learning system that improves alert quality over time by understanding previous investigation outcomes.

Comparison with Our Project: Our project implements a similar feedback mechanism through the `decision_store.py` module:

- Analysts record outcomes ('true_positive', 'false_positive', 'missed_attack', 'benign')
- The system tracks accuracy, precision, recall, and F1 score over time
- Adaptive thresholds adjust based on accumulated feedback

Key difference: Lin et al. use GenAI to suppress alerts (reduce false positives after detection). Our system reduces false positives structurally through the 60/40 fusion algorithm before the AI layer even engages, then uses AI for explainability. This dual-layer approach is more robust than relying solely on GenAI suppression.

Paper 13: "AI-Augmented SOC: Next-Generation Security Operations"

- **Authors:** (Multiple authors, survey)
- **Published:** 2025–2026, MDPI / Multiple venues

Summary: This extensive survey documents the transition toward next-generation SOCs where AI agents handle multi-step tasks including asset discovery, vulnerability management, alert triage, and incident response with minimal human intervention. It identifies key requirements: real-time processing, explainability, graduated response, and human-in-the-loop safety checkpoints.

Comparison with Our Project: Our system implements many of the architectural requirements identified in this survey:

- **Real-time processing:** CloudWatch/CloudTrail monitoring with <10ms Isolation Forest inference.
- **Explainability:** Natural language reasoning via Agentic AI.
- **Graduated response:** 4-tier LOG/ALERT/RATE_LIMIT/BLOCK system.
- **Human-in-the-loop:** Feedback recording mechanism for outcome validation.
- **Safety checkpoints:** Rule-based fallback ensuring reliable operation even if AI fails.

Our project can be viewed as a proof-of-concept implementation of the next-gen SOC architecture this survey describes.

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3. Novelty Statement

Based on the comprehensive literature review of 13 papers from 2025–2026, the **novel contributions** of this project are:

Novelty 1: Cross-Domain Weighted Fusion (IDS + UEBA)

No existing paper (including Papers 5, 7, 9, 10) combines Network IDS and User UEBA outputs using a weighted risk fusion formula as the primary detection mechanism. Existing hybrid IDS papers fuse multiple ML algorithms (Paper 9), multiple processing stages (Paper 7), or signature + anomaly methods (Paper 10). Our 60/40 Network-User domain-level fusion is architecturally unique and provides contextual correlation capabilities none of these approaches offer.

Novelty 2: End-to-End ReAct Pipeline for Cloud Threat Detection

Existing ReAct cybersecurity papers (Papers 1, 3) focus on either post-incident response or threat intelligence analysis, not the full detection-to-response pipeline. Our project is the first to implement a complete pipeline: AWS CloudWatch/CloudTrail data → dual Isolation Forest detection → 60/40 fusion → ReAct AI reasoning with tool-calling → autonomous AWS Security Group response.

Novelty 3: Zero-Cost, Privacy-Preserving AI Security Agent

Every LLM-based cybersecurity paper in this survey (Papers 1, 3, 4) relies on cloud-hosted models or frameworks (LangGraph, RAG pipelines) that incur API costs and

require transmitting security data externally. Our project achieves comparable reasoning quality using entirely local Ollama inference (Qwen 2.5), with zero cloud API costs and complete data privacy — a critical differentiator for organizations with strict data sovereignty requirements.

Novelty 4: Hybrid Mathematical + AI Detection Architecture

No paper in this survey combines a mathematical detection backbone with an AI reasoning overlay. Papers either use pure ML detection (Papers 5, 6, 8, 9) or pure AI reasoning (Papers 1, 3). Our architecture uniquely layers AI explainability on top of mathematical detection, with automatic fallback to the math if the AI fails. This provides both reliability (mathematical guarantees) and interpretability (natural language explanations).

Novelty 5: Natural Language Explainability vs. Post-Hoc XAI

The dominant XAI approach in 2025 literature is post-hoc SHAP/LIME analysis (Paper 11). Our system generates natural language explanations during the decision process as part of the ReAct reasoning trace, providing real-time, human-readable justifications that are more accessible to non-technical stakeholders than mathematical feature importance scores.

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4. Summary Comparison Table

Aspect	Current Literature (2025–2026)	Our Project
Detection	Single-modality (IDS or UEBA alone)	Dual-modality: IDS + UEBA with 60/40 fusion
Algorithm	IF + VAE/AE (Paper 5), IF + Suricata (Paper 10)	Isolation Forest x 2, fused at risk-score level
AI Reasoning	Cloud-hosted LLMs, LangGraph (Papers 1, 3)	Local Ollama ReAct agent (zero cost, full privacy)
Explainability	Post-hoc SHAP/LIME (Paper 11)	Real-time natural language reasoning traces
Autonomous Response	Framework-level (Papers 1, 4)	Implemented: LOG → ALERT → RATE_LIMIT → BLOCK
Memory & Learning	Stateless (most papers)	SQLite decision store with feedback loop
False Positive Handling	GenAI suppression (Paper 12)	Structural reduction via fusion + AI layer
Cost	Cloud API fees (Papers 1, 3)	$0 (100% local inference)
Privacy	Data sent to external APIs	Data never leaves the machine
Concept Drift	Sliding window + T-test (Paper 8)	Feedback-driven adaptive thresholds

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5. References

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